Quantitative Comparisons

ANSWER KEY

Section 1: Arithmetic & Fractional Comparisons

1 Compare: Column A: $\frac{7}{8}$, Column B: $\frac{8}{9}$.

a. **Value in Column B is greater ✓**

b. Value in Column A is greater

c. The two values are equal

d. The information given is insufficient

Cross multiply: $7 \times 9 = 63$ versus $8 \times 8 = 64$. Since $64 > 63$, $\frac{8}{9} > \frac{7}{8}$. Column B is greater.

2 Compare: Column A: $\sqrt{0.49}$, Column B: 0.7.

a. **The two values are equal ✓**

b. Value in Column A is greater

c. Value in Column B is greater

d. The information given is insufficient

$\sqrt{0.49} = \sqrt{\frac{49}{100}} = \frac{7}{10} = 0.7$. The two values are equal.

3 Compare: Column A: $\frac{1}{3} + \frac{1}{4}$, Column B: $\frac{7}{12}$.

a. **The two values are equal ✓**

b. Value in Column A is greater

c. Value in Column B is greater

d. The information given is insufficient

$\frac{1}{3} + \frac{1}{4} = \frac{4+3}{12} = \frac{7}{12}$. Both columns are identical.

Section 2: Algebraic Variables & Unknown Signs

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4 Given that $x > 0$ and $y < 0$, compare: Column A: $x - y$, Column B: $y - x$.

a. **Value in Column A is greater ✓**

b. Value in Column B is greater

c. The two values are equal

d. The information given is insufficient

Since $x > 0$ and $y < 0$, $x - y$ is positive minus negative, which is strictly positive (> 0). Conversely, $y - x$ is negative minus positive, which is strictly negative (< 0). Thus, Column A is greater.

5 Given that $a^2 = 25$, compare: Column A: a, Column B: 0.

a. **The information given is insufficient ✓**

b. Value in Column A is greater

c. Value in Column B is greater

d. The two values are equal

$a^2 = 25 \Rightarrow a = 5$ or $a = -5$. If $a = 5$, Column A is greater. If $a = -5$, Column B is greater. The relationship cannot be determined.

6 Given that $x > 1$, compare: Column A: x^2, Column B: x^3.

a. **Value in Column B is greater ✓**

b. Value in Column A is greater

c. The two values are equal

d. The information given is insufficient

For any real number $x > 1$, higher powers are strictly larger: $x^3 > x^2$. Column B is greater.

Section 3: Geometric & Measurement Comparisons

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7 A circle has radius 5 cm and a square has side length 9 cm. Compare: Column A: Area of the circle, Column B: Area of the square.

a. **Value in Column B is greater ✓**

b. Value in Column A is greater

c. The two values are equal

d. The information given is insufficient

Circle area $= \pi(5^2) = 25\pi \approx 25 \times 3.14 = 78.5$ cm^2. Square area $= 9^2 = 81$ cm^2. Since $81 > 78.5$, Column B is greater.

8 In $\triangle ABC$, $\angle A = 50^\circ$ and $\angle B = 70^\circ$. Compare: Column A: Length of side AB, Column B: Length of side AC.

a. **Value in Column B is greater ✓**

b. Value in Column A is greater

c. The two values are equal

d. The information given is insufficient

Third angle $\angle C = 180 - (50 + 70) = 60^\circ$. Side AB is opposite $\angle C = 60^\circ$. Side AC is opposite $\angle B = 70^\circ$. In any triangle, larger angles face longer sides. Thus, $AC > AB$.

9 Compare: Column A: The perimeter of a rectangle with length 8 and width 4, Column B: The perimeter of a square of area 36.

a. **The two values are equal ✓**

b. Value in Column A is greater

c. Value in Column B is greater

d. The information given is insufficient

Rectangle perimeter $= 2(8 + 4) = 24$. Square side $= \sqrt{36} = 6$, and its perimeter $= 4 \times 6 = 24$. The two values are equal.

Section 4: Powers, Radicals & Number Theory

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10 Compare: Column A: 2^{50}, Column B: 3^{30}.

a. **Value in Column A is greater ✓**

b. Value in Column B is greater

c. The two values are equal

d. The information given is insufficient

Express with a common power of 10: $2^{50} = (2^5)^{10} = 32^{10}$. $3^{30} = (3^3)^{10} = 27^{10}$. Since $32 > 27$, $32^{10} > 27^{10}$, so Column A is greater.

11 Compare: Column A: $\sqrt{64 + 36}$, Column B: $\sqrt{64} + \sqrt{36}$.

a. **Value in Column B is greater ✓**

b. Value in Column A is greater

c. The two values are equal

d. The information given is insufficient

Column A: $\sqrt{100} = 10$. Column B: $8 + 6 = 14$. Since $14 > 10$, Column B is greater.

12 Compare: Column A: $(-1)^{100}$, Column B: $(-1)^{101}$.

a. **Value in Column A is greater ✓**

b. Value in Column B is greater

c. The two values are equal

d. The information given is insufficient

Column A: Even power $(-1)^{100} = 1$. Column B: Odd power $(-1)^{101} = -1$. Since $1 > -1$, Column A is greater.

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